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Bacterial Diversity and Bioprospecting for Cold-Active Lipases, Amylases and Proteases, from Culturable Bacteria of Kongsfjorden and Ny-Ålesund, Svalbard, Arctic

T. N. R. Srinivas · S. S. S. Nageswara Rao · P. Vishnu Vardhan Reddy · M. S. Pratibha · B. Sailaja · B. Kavya · K. Hara Kishore · Z. Begum · S. M. Singh · S. Shivaji

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Abstract Culturable bacterial diversity of seven marine sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, Svalbard, Arctic was studied. The bacterial abundance in the marine sediments of Kongsfjorden varied marginally (0.5×10^3–1.3×10^4 cfu/g sediment) and the bacterial number in the two samples collected from the shore of Ny-Ålesund also was very similar (0.6×10^4 and 3.4×10^4, respectively). From the nine samples a total of 103 bacterial isolates were obtained and these isolates could be grouped in to 47 phylotypes based on the 16S rRNA gene sequence belonging to 4 phyla namely *Actinobacteria, Bacilli, Bacteroidetes* and *Proteobacteria*. Representatives of the 47 phylotypes varied in their growth temperature range (4–37°C), in their tolerance to NaCl (0.3–2 M NaCl) and growth pH range (2–11). Representatives of 26 phylotypes exhibited amylase and lipase activity either at 5 or 20°C or at both the temperatures. A few of the representatives exhibited amylase and/or lipase activity only at 5°C. None of the phylotypes exhibited protease activity. Most of the phylotypes (38) were pigmented. Fatty acid profile studies indicated that short chain fatty acids, unsaturated fatty acids, branched fatty acids, the cyclic and the *cis* fatty acids are predominant in the psychrophilic bacteria.

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T. N. R. Srinivas · S. S. S. Nageswara Rao · P. Vishnu Vardhan Reddy · M. S. Pratibha · B. Sailaja · B. Kavya · K. Hara Kishore · Z. Begum · S. Shivaji (✉)
Centre for Cellular and Molecular Biology, Uppal Road, Hyderabad 500-007, India
e-mail: shivas@ccmb.res.in

T. N. R. Srinivas
e-mail: msriv@ccmb.res.in

S. S. S. Nageswara Rao
e-mail: shivagenetech@gmail.com

P. Vishnu Vardhan Reddy
e-mail: pvishnu@ccmb.res.in

M. S. Pratibha
e-mail: msprati@ccmb.res.in

B. Sailaja
e-mail: bsailaja@ccmb.res.in

B. Kavya
e-mail: nature_kavya@yahoo.com

K. Hara Kishore
e-mail: hara_kishore@yahoo.co.in

Z. Begum
e-mail: zareena@ccmb.res.in

S. M. Singh
National Centre for Antarctic and Ocean Research, Headland Sada, Vasco da Gama, Goa, India
e-mail: drsmsingh@yahoo.com

Introduction

The two Arctic fjords Kongsfjorden and Krossfjorden located off the west coast of Spitsbergen have attracted the attention of scientists as a model site for studies on the impact of climate change in the Arctic [38] and they are regarded as key European sites for Arctic biodiversity monitoring [13]. Earlier studies in these fjords have focused both on the physical characteristics of the environment and the biotic components of the ecosystem [13, 31]. Benthic

bacteria have received little attention in Kongsfjord, but have been examined in some of the other fjords in Svalbard and on the west coast shelf [5, 18, 26, 27].

Despite the fact that Arctic has a number of fjords comparatively little is known about the bacterial diversity of the fjords [16, 27]. In a recent study, Jankowska et al. [16] studied the bacterial abundance and biomass in water samples from Krossfjorden and Kongsfjorden. However, no attempts were made to either identify the various bacteria in the study or bioprospect for cold-active enzymes, important for biotechnology processes [9]. The present study is an attempt on the above lines using sediment samples from different regions of the Kongsfjorden.

Materials and Methods

Sampling Site

Kongsfjorden (79° N, 12° E) is about 25 km in length (Fig. 1) and is fed by inflows of water from Kongsbreen, the most active glacier in the Svalbard Archipelago [21]. Each year the glacier inputs 2.6×10^5 m^3 of mineral material into fjord waters per season [8]. Thus, the inflow of water and minerals causes unique temperature and salinity zones such that the surface water layer is of relatively high temperature and very low salinity compared to the lower water layer [32].

Sampling Method

Seven sediment samples (Table 1) were collected on 9th to 18th of August 2007, in sterile 50 ml Falcon tubes, starting from the innermost point of the Kongsfjord to the outer most point (Table 1; Fig. 1). The sediment samples were retrieved using a grab sampler and only the middle portion of the sample was transferred into collection tubes using a sterile spatula. In addition one soil (SH) and one sediment (marine) sample were also collected near the shore of Ny-Ålesund. The mean air temperature at the time of collection was 4.5°C. All the samples were stored at 5°C and transported to the Centre for Cellular and Molecular Biology, Hyderabad, India, for further processing.

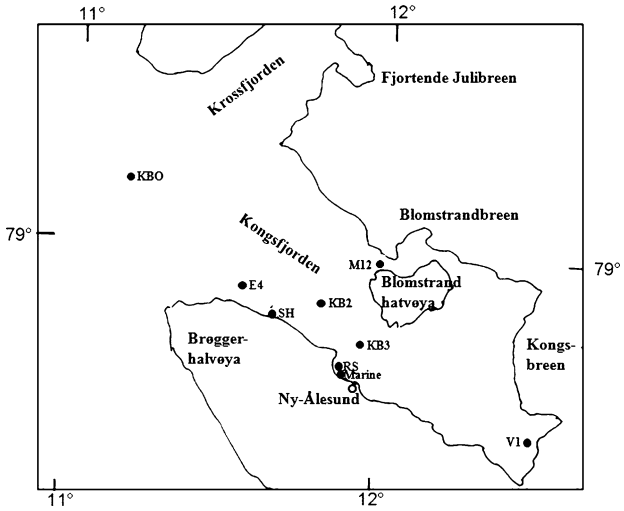

Fig. 1 Location of sampling sites in Kongsfjorden and Ny-Ålesund Svalbard, Arctic

Isolation and Characterisation of Bacterial Strains

Using 100 mg of the sediment sample bacteria were isolated as described earlier [30] using Zobell marine agar plates incubated at 10°C for 15 days. Colony characteristics and growth at different temperatures, pH and salt concentrations were carried out on Zobell marine agar plates. Extracellular enzymatic activities like amylase [25], lipase [6] and protease [25] were checked by streaking the culture on Zobell marine agar plates supplemented with either 0.2% soluble starch; 1% Tween-60 along with 0.01% $CaCl_2$; and 0.3% casein, respectively, and incubating the plates at 5 and 20°C for 5–10 days. Fatty acid methyl esters were prepared and analysed as previously described [31] using cells grown at 20°C. For pigment extraction, 2 ml of log phase culture was centrifuged, supernatant discarded, 2 ml of acetone or ethanol added and mixed thoroughly on a vortex mixer, centrifuged and supernatant transferred to a new centrifuge tube. The absorption spectrum from 300 to 800 nm was measured on a PerkinElmer Lambda 35 spectrometer (Massachusetts 02451, USA) using either acetone or ethanol as a blank.

PCR Amplification of the 16S rRNA Gene, Sequencing and Phylogenetic Analysis

The 16S rRNA gene was amplified and sequenced as previously described [29] using conserved primers [20]. 16S rRNA gene sequences were subjected to BLAST [2] and aligned using the CLUSTAL_X [33]. Phylogenetic analysis was initially done using the 16S rRNA gene sequences of all the 103 strains. Subsequently 47 strains representing different taxa were selected. Phylogenetic trees were constructed using Maximum likelihood (ML) and Neighbor-joining (NJ) methods [4]. All the 16S rRNA gene sequences of the strains were deposited in GenBank with accession numbers FN377697–FN377708, FN377710–FN377732 and FN377734–FN377745.

Statistical Analysis

In order to compare the bacterial diversity within the nine samples, 16S rRNA gene sequences of the isolates showing

Table 1 Bacterial number (cfu/g) in sediment samples of Kongsfjorden region of Svalbard, Arctic

Serial number	Sample number	GPS position	Water surface/bottom temperature (°C)	pH	Depth (m)	Description of sediment	CFU/g	Number of isolates	Strain numbers
1	V1	78° 53′ 537″ N 12° 28′ 300″ E	4.2/1.5	7.75	200	Brownish, sticky, sparse vegetation and polychaete worms visible	0.5×10^3	20	VI-1; VI-2; V1-3; VI-4A; V1-4B; V1-6; V1-6A; V1-8, V1-11; VI-16; VI-22A; VI-22B; V1-41; VI-55; VI-71, VI-91; VI-92; VI-93; V1-99A; VI-99B
2	KB3	78° 57′ 179″ N 11° 59′ 567″ E	5.0/2.5	7.80	330	Black on the surface and dark brown inside. Clay like, sticky with cylindrical plants/weeds on surface	1.0×10^4	8	KB3-1; KB3-2; KB3-3; KB3-11; KB3-12; KB3-13; KB3-14; KB3-15
3	KB2	78° 53′ 427″ N 11° 45′ 223″ E	5.1/4.2	7.80	295	Black, clay like, sticky with very few cylindrical plants/weeds on surface	1.0×10^4	5	KB2-1; KB2-3; KB2-4; KB2-6; KB2-7
4	M12	78° 59′ 725″ N 11° 56′ 100″ E	4.7–/4.0	7.90	200	Black on the surface and dark brown inside. Clay like, sticky with cylindrical plants/weeds on surface, polychaete worms visible	1.0×10^4	8	M12-1; M12-3; M12-4; M12-11A; M12-11B; M12-90; M12-91A; M12-91B
5	E4	78° 59′ 091″ N 11° 31′ 731″ E	4.8/2.7	7.9	276	Black, clay like, sticky, with pebbles and vegetation, polychaete worms	0.8×10^4	12	E4-1; E4-2; E4-3A; E4-3B; E4-4; E4-5; E4-6; E4-7; E4-8; E4-9A; E4-9B; E4-10
6	KB0	79° 02′ 554″ N 11° 07′ 851″ E	6.6/3.3	7.6	310.5	Black clay like, with pebbles sparse vegetation	1.3×10^4	5	KBO-1; KBO-2; KBO-3; KBO-4; KBO-5
7	RS	78° 56′ 079″ N 11° 52′ 549″ E	5.2/5.2	7.60	4.2	Brown, clay like, sticky with algae	0.8×10^4	16	RS-1; RS-2; RS-3; RS-4; RS-5; RS-7; RS-8; RS-9; RS-21; RS-32; RS-33; RS-42; RS-61A; RS-61B; RS-62; RS-63
8	SH	78° 57′ 85″ N 11° 39′ 36″ E	–/5.5	7.8	0	Dry soil sample from the coast of Ny-Ålesund	0.6×10^4	21	SH-1; SH-10A; SH-10B; SH-11; SH-22; SH-31; SH-32; SH-41; SH-43A; SH-43B; SH-44; SH-61A; SH-63; SH-82A; SH-82B; SH-93,5A; SH-93,5B; SH-94A; SH-94B; SH-96A; SH-96B
9	Marine	78° 55′ 699″ N 11° 53′ 121″ E	8.8/5.5	7.8	0.5	Soil sample from the coast of Ny-Ålesund	3.4×10^4	8	Marine-1; Marine-4; Marine-5; Marine-9; Marine-13; Marine-15; Marine-22; Marine-33

≥97% sequence similarity were grouped into the same OTU (phylotype). Three different softwares namely Shannon–Wiener Diversity Index/Shannon Entropy Calculator (http://www.changbioscience.com/genetics/shannon.html); Rarefaction Calculator (http://www2.biology.ualberta.ca/jbrzusto/rarefact.php) and Online Calculation (http://biome.sdsu.edu/fastgroup/cal_tools.htm) were used to calculate Shannon index (H′), Evenness (J′) [21] and the Simpson's index (*D*) [22]. Coverage of 16S rRNA gene sequences of the isolates was calculated as described previously [11]. Rarefaction curves were generated to compare the relative diversity and coverage of each.

Results

Sediment Description

Seven sediment samples were collected from different parts of Kongsfjorden and the samples were black to brownish in colour, clay like, sticky and contained fragmented pieces of vegetation which was brown to black in colour with a diameter of less than 10 mm and length varying from 5 to 15 cm embedded in the sediment. Pebbles and polychaete worms were visible in some of the samples (Table 1).

Viable Bacterial Counts

A total of 103 bacterial strains were recovered from the 9 samples listed in Table 1. The bacterial abundance in the 7 sediments from the innermost point of the Kongsfjorden (V1) to the outermost point (KBO) varied from 0.5×10^3 to 1.3×10^4 cfu/g sediment. Further, the nature of the sediment with respect to whether it was hard, soft and with or without pebbles or vegetation did not appear to influence the bacterial abundance (Table 1). The bacterial number in the two samples (SH and Marine) collected from the shore of Ny-Ålesund also was very similar (0.6×10^4 and 3.4×10^4, respectively) (Table 1).

Statistical Analysis

The 103 strains from the 9 sampling sites based on similarity criteria of >97% at the 16S rRNA gene sequence could be categorised into 47 phylotypes (Table 2; Supplementary Table 1). Out of the 47 phylotypes only 8 phylotypes were common to two or more samples and the most common phylotype was *Psychrobacter nivimaris*, which was shared by 5 out of the 7 sediment samples. Species richness varied from 2 to 11 and the diversity coverage ranged from 12.5 to 60% (Table 3). The individual rarefaction curves for all the 9 samples indicated that the bacterial populations were the least diverse in KB2 and most diverse in V1 but the rarefaction curve did not plateau at any of the sites except for KB2, with diversity coverage of 60%, where the curve showed a tendency to plateau (Fig. 2). The rarefaction curve analysis implies that these are likely to be minimal estimates of diversity. These observations are supported by bacterial diversity parameters, such as Shannon index, Simpson's index and Evenness (Table 3).

Identification of the Isolated Strains

Phylogenetic analysis of all the 16S rRNA gene sequences indicated that the 103 strains could be minimised into 47 phylotypes, each phylotypes representing a different taxon. Sequence similarity of representative of each phylotype compared to the nearest phylogenetic neighbour ranged from 94 to 99 and one of the strains showed 100% similarity with the nearest phylogenetic neighbour (Table 2). Representative strains of the 47 phylotypes were selected for subsequent studies.

Taxonomic analysis of the representative strains of each of the 47 phylotype indicated that 38 groups were represented by Gram-negative strains and 9 by Gram-positive strains. The phylogenetic trees constructed to determine their affiliations are shown in Figs. 3a and b. Out of the 38 Gram-negative phylotypes 32 belonged to the phylum *Proteobacteria*, five to the class *Flavobacteria* and one to class *Sphingobacteria* (Table 2). Among the *Proteobacteria,* species of *Pseudomonas* and *Pseudoalteromonas* appeared to be the most predominant genera (Table 2). Out of the 9 Gram-positive phylotypes, 8 were affiliated to the class *Actinobacteria* and were represented by various species of *Arthrobacter*, *Rhodococcus qingshengii* and *Salinibacterium amurskyense* (Table 2). All phylogenetic neighbours of the 8 phylotypes affiliated to *Actinobacteria* were isolated from cold habitats and reported to be psychrophilic. Based on the sequence similarity results four strains (E4-1; E4-4; RS-3 and SH-94B) are likely to be novel species as they differ with the nearest phylogenetic neighbour at the 16S rRNA gene sequence level by 3–7.5% (Table 2).

Colony, Growth and Pigment Characteristics of the Representative Strains

The representative strains of the 47 phylotypes showed colony morphology similar to that reported for the nearest phylogenetic neighbour (data not shown). Representatives of 16 phylotypes were psychrophilic and could grow in the temperature range of 4–20°C but not above 20°C. Further, 26 strains could grow between 4 and 30°C and 5 strains

Table 2 Identification of the 103 bacterial isolates from seven sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, Svalbard, Arctic based on phylogenetic analysis of the 16S rRNA gene sequences

Serial number	Strain number	Nearest phylogenetic relative	16S rRNA gene sequence similarity (%)
Proteobacteria			
Alphaproteobacteria			
1	**Marine-1**	*Brevundimonas vesicularis* LMG 2350^T (AJ227780)	100.0
2	**SH-1**	*Paracoccus marcusii* DSM 11574^T (Y12703)	99.7
3	**E4-1**	*Roseovarius aestuarii* SMK-122^T (EU156066)	92.5
4	**V1-2**	*Sphingomonas panni* C52^T (AJ575818)	98.1
5	**KBO-5**	*Sphingopyxis flavimaris* SW-151^T (AY554010)	99.1
6	**M12-90**	*Sulfitobacter litoralis* Iso 3^T (DQ097527)	99.7
Gammaproteobacteria			
7	**SH-94B**	*Acinetobacter radioresistens* DSM 6976^T (X81666)	95.6
8	**E4-4**	*Colwellia aestuarii* SMK-10^T (DQ055844)	97
9	**KB3-12**	*Enhydrobacter aerosaccus* LMG 21877^T (AJ550856)	99.7
10	V1-4A; V1-6; V1-6A; **V1-8**; V1-99A; V1-99B	*Marinobacter maritimus* CK47^T (AJ704395)	97.8
11	**RS-21**	*Marinobacterium rhizophilum* CL-YJ9^T (EF192391)	97.5
12	**M12-1**	*Marinomonas primoryensis* KMM 3633^T (AB074193)	99.4
13	V1-11; **V1-41**; V1-55; V1-92	*Oceanisphaera donghaensis* BL1^T (DQ190441)	97.8
14	**KB2-7**	*Photobacterium frigidiphilum* SL13^T (AY538749)	99.7
15	KB3-3; **M12-11A**; M12-91A	*Pseudoalteromonas arctica* A 37-1-2^T (DQ787199))	99.4
16	E4-3A; E4-3B; KBO-2; KBO-4; KB2-1; KB2-3; KB2-4; KB2-6; KB3-1; **KB3-13**; M12-91B	*Pseudoalteromonas elyakovii* KMM 162^T (AF082562)	98.4
17	**M12-4**	*Pseudoalteromonas nigrifaciens* NCIMB 8614^T (X82146)	99.7
18	**M12-3**	*Pseudoalteromonas paragorgicola* KMM 3548^T (AY040229)	99.7
19	**RS-61A**; RS-61B	*Pseudoalteromonas prydzensis* MB8-11^T (U85855)	98.1
20	**RS-32**	*Pseudoalteromonas tetraodonis* IAM 14160^T (AF214730)	98.7
21	V1-4B; **KBO-3**; KB3-15	*Pseudoalteromonas translucida* KMM 520^T (AY040230	98.1
22	**E4-5**	*Pseudomonas azotoformans* IAM1603^T (D84009)	99.4
23	V1-16; **V1-22B**	*Pseudomonas brenneri* CFML 97-391^T (AF268968)	97.8
24	RS-1; RS-2; RS-4; RS-8; **RS-42**	*Pseudomonas guineae* M8^T (AM491810)	99.4
25	Marine-5; **Marine-13**	*Pseudomonas mandelii* CIP 105273^T (AF058286)	98.4
26	**V1-1**	*Pseudomonas simiae* OLi^T (AJ936933)	98.7
27	**V1-71**	*Pseudomonas stutzeri* CCUG 11256^T (U26262)	100.0
28	**V1-91**	*Pseudomonas trivialis* DSM 14937^T (AJ492831)	98.7
29	**KBO-1**; KB3-11	*Psychrobacter aquimaris* SW-210^T (AY722804)	99.1
30	**KB3-14**	*Psychrobacter fozii* NF23^T (AJ430827)	99.7

Table 2 continued

Serial number	Strain number	Nearest phylogenetic relative	16S rRNA gene sequence similarity (%)
31	E4-2; **E4-10**; V1-93; KB3-2	*Psychrobacter nivimaris* 88/2-7^T (AJ313425)	98.7
32	**RS-5**; RS-7; RS-9; RS-62; RS-63	*Shewanella livingstonensis* LMG 19866^T (AJ300834)	98.7
Flavobacteria			
33	**V1-3**	*Flavobacterium degerlachei* LMG 21915^T (AJ557886)	98.1
34	**E4-9a**	*Lacinutrix copepodicola* DJ3^T (AY694001)	98.1
35	**E4-6**; E4-7	*Maribacter orientalis* KMM 3947^T (AY271624)	98.1
36	**RS-3**	*Winogradskyella thalassocola* KMM 3907^T (AY521223)	97.0
37	E4-8; **E4-9b**	*Zobellia laminariae* KMM 3676^T (AB121975)	99.4
Sphingobacteria			
38	**M12-11B**	*Cyclobacterium amurskyense* KMM 6143^T (AY960985)	97.8
Actinobacteria			
39	SH-43A; **SH-43B**; SH-44	*Arthrobacter gangotriensis* Lz1y^T (AJ606061)	97.3
40	Marine-9; Marine-22; **RS-33**	*Arthrobacter oryzae* KV-651^T (AB279889)	97.5
41	Marine-4; **Marine-33**; SH-11; SH-22; SH-32; SH-41; SH-63	*Arthrobacter polychromogenes* DSM 20136^T (X80741)	99.0
42	**SH-96A**	*Arthrobacter psychrophenolicus* AG31^T (AJ616763)	98.5
43	**SH-82B**	*Arthrobacter stackebrandtii* CCM 2783^T (AJ640198)	98.0
44	Marine-15; SH-10A; SH-10B; **SH-93,5B**	*Arthrobacter sulfureus* DSM 20167^T (X83409)	97.8
45	**VI-22A**	*Rhodococcus qingshengii* djl-6^T (DQ090961)	100
46	**SH-61A**	*Salinibacterium amurskyense* KMM 3673^T (AF539697)	98.8
Bacilli			
47	SH-31; SH-82A; SH-93,5A; SH-94A; **SH-96B**	*Planococcus kocurii* NCIMB 629^T (X62173)	98.5

The almost complete 16S rRNA gene sequences (956–1547 bases) of all the 103 strains was obtained and subjected to BLAST and phylogenetic analysis. Based on the phylogenetic analysis the 103 strains were categorised in to 47 groups and representative sequence (956–1547 bases) of each group was submitted to EMBL. The accession numbers of the grouped strains (indicated in *bold*) are FN377697–FN377708, FN377710–FN377732 and FN377734–FN377745

Table 3 Diversity indices for the isolates from the seven marine sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, Svalbard, Arctic

	V1	KB3	M12	RS	KB2	SH	E4	KB0	Marine
Number of isolates	20	8	8	16	5	21	12	5	8
Species richness	11	7	7	8	2	9	8	4	5
Coverage (%)	45	12.5	12.5	53	60	55	33	20	37.5
Evenness (J′)	0.88	0.98	0.98	0.87	0.72	0.89	0.97	0.96	0.97
Shannon (H′)	2.11	1.90	1.90	1.80	0.50	2.06	2.02	1.33	1.56
Simpson's (*D*)	0.16	0.15	0.15	0.20	0.68	0.15	0.14	0.28	0.22

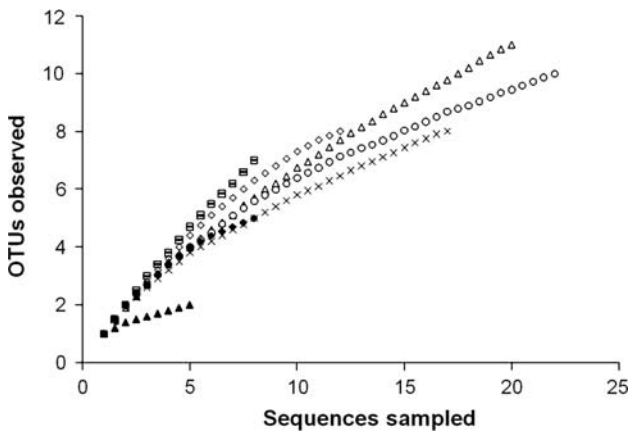

Fig. 2 Rarefaction curves of observed OTUs in the seven marine sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund. The symbols represent the rarefaction curves generated for the isolates obtained from samples V1 (*open triangle*), KB3 (*open square*), M12 (*minus*), RS (*times*), KB2 (*filled triangle*), SH (*open circle*), E4 (*open diamond*), KBO (*plus*) and marine (*filled diamond*)

between 4 and 37°C indicating that they are psychrotolerant (Table 4). All the stains could grow without NaCl in the medium and the tolerance to NaCl varied from 0.3 to 2 M (Table 4). Strains could grow in the pH range of 2–11 and few strains could grow only in a narrow range of 5–8 and 6–9 (Table 4).

Attempts were made to extract the pigment from 23 of the 38 representative strains of the phylotypes for which either the colony colour was reported or the pigment was characterised at least partially from the nearest phylogenetic neighbour (Supplementary Table 2). Only in 12 strains a carotenoid type of pigment was identified based on the multiple absorption maxima (Supplementary Table 2).

In the present study, only 26 strains (56%) out of 47 strains showed either amylase or lipase or both activities at 5°C and (or) 20°C (Table 4). Only one strain RS-32 showed both amylase and lipase activity only at 5°C indicating that it could be a good source for cold-active and heat labile amylase and lipase, as are strains V1-1, V1-22B, KB2-7, E4-10 and KBO-1 (Table 4).

Fatty Acid Profiles

In the present study, short chain fatty acids were observed in most Gram-negative bacteria but they were not so predominant in the Gram-positive bacteria (Supplementary Tables 3, 4). The results also clearly indicate that in the Gram-negative and Gram-positive bacteria iso-, anteiso-, methylated-, cyclo-, hydroxyl- and mono- unsaturated fatty acids together constitute a significant proportion of the total fatty acid composition varying from a minimum of 16.7% to a maximum of 100% as in SH-43B and SH-96A, respectively.

Discussion

The present study was mainly directed towards identifying the viable bacteria from the marine sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, Svalbard, Arctic and to bioprospect for enzymes from the isolates. Bacterial viable counts varied only marginally (0.5×10^3–1.3×10^4 cfu/g sediment) in the 7 sediments from the innermost point of the Kongsfjorden (V1) to the outermost point (KBO). The sediment samples, varied in grain size from clay like from inner fjord to predominantly rock at the outer Kongsfjorden, was in black to brown in colour, had vegetation and polychaete worms [17, 19, 37]. It was difficult to identify the plant material in the sediment but similar plant material from the same fjord were identified based on morphology as belonging to *Chlorophyta* [37]. The black colour of the sediment may indicate its anoxic nature and anoxic processes are known to significantly influence organic matter remineralization in Svalbard fjords [19]. The carbon content of the sediment was not determined but it is known to vary from a minimum of 0.0824 g C m^{-2} d^{-1} in winter to a maximum of 9.2227 g C m^{-2} d^{-1} in summer [13].

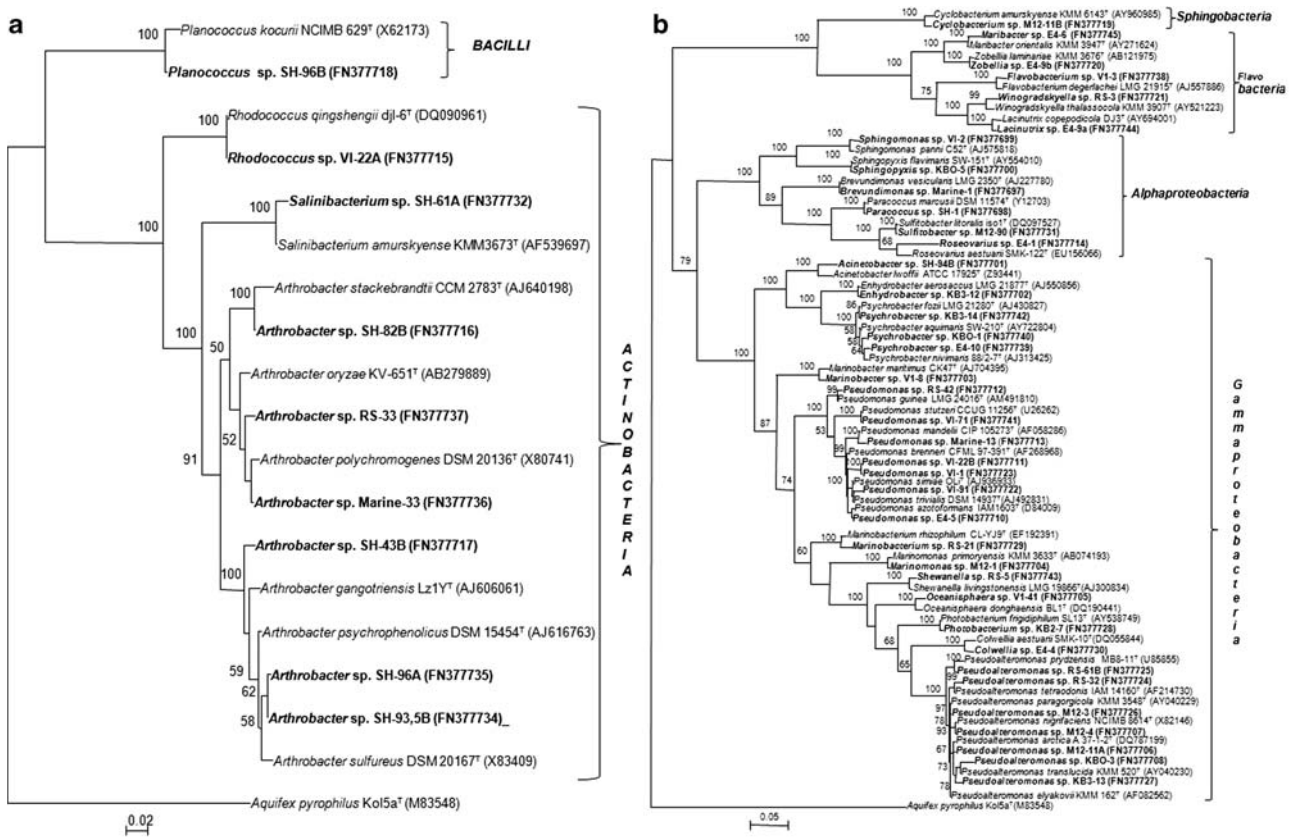

Fig. 3 Phylogenetic trees based on 16S rRNA gene sequences showing the relationship of Gram-positive (**a**) and Gram-negative strains (**b**) obtained from the seven marine sediment samples collected from Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, with their nearest phylogenetic relatives. Phylogenetic trees were constructed by Maximum likelihood method. Numbers at nodes are bootstrap values. The bar represents 0.02 and 0.05 substitutions per alignment position in **a** and **b**, respectively

To the best of our knowledge, this is the first report on the viable bacterial abundance in sediments of Kongsfjorden though Jankowska et al. [16] showed that the total bacterial number in the water of Kongsfjorden and Krossfjorden, using acridine orange, ranged from 7.8 to $80 \times 10^5/cm^3$ of water. The viable count reported in the present paper, are underestimated since a single medium only, Zobell marine agar [39] was used which, however, has been widely used for enumeration of viable heterotrophic bacteria in the marine environments [36, 39]. Zobell marine agar enriched with glucose, starch, $FeSO_4$ and Na_2HPO_4 gave significantly higher total viable counts [34] compared to Zobell marine agar.

Based on similarity threshold value of >97% 16S rRNA gene sequence similarities the 103 strains from the 9 sampling sites were categorised into 47 phylotypes (Table 2; Supplementary Table 1). Taxonomic analysis of the 47 representative strains of each phylotype indicated that 38 phylotypes were represented by Gram-negative strains and 9 by Gram-positive strains. Phylogenetic neighbours of the strains representing the 38 phylotypes affiliated to *Proteobacteria, Flavobacteria* and *Sphingobacteria* have been isolated earlier from cold habitats [1, 35] and reported to be psychrophilic except *Enhydrobacter aerosaccus*, *Marinobacterium rhizophilum* and *Oceanisphaera donghaensis*. One of the Gram-positive phylotype was affiliated to class *Bacilli* and the nearest phylogenetic neighbour was *Planococcus kocurii* isolated from the frozen foods [12]. It is well-documented that the most frequently encountered bacterial genera in cold environments belong to *Proteobacteria*, *Cytophaga-Flavobacterium-Bacteroidetes* (CFB) group and low and high G + C Gram-positive genera [3]. The presence of these strains in these cold environments suggests that their tolerant to cold conditions and this may be attributed to the presence of pigments or polyunsaturated fatty acids which have been implicated in cold adaptation [23, 31].

All the Arctic isolates were capable of growing between 4 and 37°C thus indicating that psychrotolerant bacteria are predominant in the sediment of Kongsfjorden thus confirming an earlier report [28]. Further, pigmented strains are frequently found in cold environments [23] and

Table 4 Phenotypic characteristics and enzymatic activities of the representative strains of the 47 phylotypes isolated from seven marine sediment samples of Kongsfjorden and a sediment and a soil sample from Ny-Ålesund, Svalbard, Arctic

Serial number	Strain number	Phenotypic characteristics			Enzyme activities*			
		Temp. range (°C)	NaCl tolerance (M)	pH range	Amylase		Lipase	
					5°C	20°C	5°C	20°C
1	V1-1	4–30	0.5	5–11	+	+	+++	−
2	V1-2	4–20	1	7–11	−	−	−	−
3	V1-3	4–30	0.5	9–11	−	−	−	−
4	V1-8	4–30	2	6–9	+	+	++	++
5	VI-22A	4–37	0.5	5–9	+	+	−	−
6	V1-22B	4–20	0.5	5–11	+	+	+++	−
7	V1-41	4–37	0.5	7–11	−	+	−	−
8	VI-71	4–37	0.5	2–9	−	−	−	−
9	VI-91	4–30	2	5–9	−	+	−	−
10	KB3-12	4–30	2	7–11	+	−	+++	+++
11	KB3-13	4–30	1	6–9	−	−	−	−
12	KB3-14	4–20	2	7–11	−	−	+++	+++
13	M12-1	4–30	0.5	6–9	−	−	+++	+++
14	M12-3	4–20	1	7–11	+	+	−	−
15	M12-4	4–30	1.5	7–11	+	+	+++	+++
16	M12-11A	4–30	1.5	6–9	+	+	−	−
17	M12-11B	4–30	1	6–9	−	−	−	−
18	M12-90	4–20	0.5	5–11	−	−	−	++
19	RS-3	4–30	1	6–9		−	−	−
20	RS-5	4–20	0.5	6–11	+	+	−	−
21	RS-21	4–20	0.5	5–8	−	−	−	−
22	RS-32	4–30	2	6–11	+	−	++	−
23	RS-33	4–20	0.5	5–8	−	−	−	−
24	RS-42	4–30	0.5	7–11	−	−	−	+
25	RS-61B	4–30	1	6–9	+	+	−	−
26	KB2-7	4–20	0.5	5–9	+	+	+++	−
27	SH-1	4–30	0.5	7–11	−	+	−	−
28	SH-43B	4–30	0.5	7–11	−	−	−	−
29	SH-61A	4–30	0.5	7–11	−	−	−	−
30	SH-82B	4–20	0.5	5–9	−	−	−	−
31	SH-93,5B	4–30	0.5	6–9	−	−	−	−
32	SH-94B	4–30	0.5	7–11	−	−	−	−
33	SH-96A	4–37	0.5	5–11	−	−	−	−
34	SH-96B	4–37	1.5	7–11	−	−	−	−
35	E4-1	4–20	1.5	7–11	−	−	−	−
36	E4-4	4–20	0.3	7–9	+	+	+	+
37	E4-5	4–20	0.5	5–11	−	−	−	−
38	E4-6	4–20	1.5	7–11	−	−	−	−
39	E4-9a	4–20	0.5	5–9	+	−	−	−
40	E4-9b	4–30	0.5	6–9	+	+	−	+
41	E4-10	4–30	2.0	5–11	−	−	+++	−
42	KBO-1	4–30	1.0	7–11	−	−	+++	−
43	KBO-3	4–30	1.0	7–11	−	−	+++	+++
44	KBO-5	4–20	0.5	6–9	−	−	−	−
45	Marine-1	4–30	0.5	6–9	−	+	−	−

Table 4 continued

Serial number	Strain number	Phenotypic characteristics			Enzyme activities*			
		Temp. range (°C)	NaCl tolerance (M)	pH range	Amylase		Lipase	
					5°C	20°C	5°C	20°C
46	Marine-13	4–30	0.5	6–11	–	–	–	–
47	Marine-33	4–30	0.5	7–11	–	–	–	–

* All the strains were negative for protease activity

+ weak positive, ++ moderately positive, +++ highly positive, – negative

pigments have been implicated in cold adaptation [10, 14, 15, 31]. Therefore, it is not surprising that 38 out of the 47 representative strains of the phylotypes were pigmented.

Psychrophilic bacteria have attracted the attention of the scientific community [9] due to their ability to produce cold-active enzymes, with potential applications in a broad range of industrial, agricultural and medical processes. When the enzymatic activities of the isolates (Table 4) were compared with those of their nearest phylogenetic neighbour (Table 2) strains E4-1, M12-11B, SH-43B, RS-33 and SH-61A, showed properties not matched by their relatives. Other strains showed similarities by exhibiting at least one or two enzyme activities. For instance Marine-1, M12-3 and RS-61B exhibited amylase activity as in the corresponding phylogenetic neighbours *Brevundimonas vesicularis*, *Pseudoalteromonas paragorgicola* and *Pseudoalteromonas prydzensis*, respectively; E4-4, V1-8, KB2-7, M12-4, RS-32, KBO-3, V1-1 and KBO-1 showed lipase activity as in *Colwellia aestuarii*, *Marinobacter maritimus*, *Photobacterium frigidiphilum*, *Pseudoalteromonas nigrifaciens*, *Pseudoalteromonas tetraodonis*, *Pseudoalteromonas translucida*, *Pseudomonas simiae*, *Psychrobacter aquimaris* and *Zobellia laminariae*, respectively. All the other strains varied with respect to their ability to exhibit enzyme activity compared to the nearest phylogenetic neighbour. For instance there are reports of amylase and (or) lipase activities in other relatives activities are neither observed at 5°C nor at 20°C. Surprisingly, none of the strains produced extracellular proteases although protease activity has been reported in 24 nearest phylogenetic neighbours [1].

It is now reasonably well-established that synthesis of unsaturated fatty acids is absolutely essential for survival at low temperatures [24]. In addition, polyunsaturated fatty acids from psychrophiles could be used as dietary supplements. With this in view the fatty acid composition of all the representative strains of the 47 phylotypes was determined. These results are in accordance with earlier studies which indicated that short chain fatty acids with <12 carbons, the unsaturated, the branched (including the iso- and anteiso- fatty acids), the cyclic and the *cis* fatty acids maintain the membrane in a fluid state and are thus known to be common in psychrophilic bacteria [7]. It has also been shown that psychrophilic bacteria as compared to mesophilic bacteria have a greater proportion of unsaturated fatty acids [31].

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References

1. Al Khudary R, Stösser NI, Qoura F et al (2008) *Pseudoalteromonas arctica* sp. nov., an aerobic, psychrotolerant, marine bacterium isolated from Spitzbergen. Int J Syst Evol Microbiol 58:2018–2024
2. Altschul SF, Gish W, Miller W et al (1990) Basic local alignment search tool. Mol Biol 215:403–410
3. Amato P, Hennebelle R, Magand O et al (2007) Bacterial characterization of the snow cover at Sptizberg, Svalbard. FEMS Microbiol Ecol 59:255–264
4. Anil Kumar P, Srinivas TNR, Madhu S et al (2009) *Indibacter alkaliphilus* gen. nov., sp. nov., a novel alkaliphilic bacterium isolated from haloalkaline Lonar Lake, India. Int J Syst Evol Microbiol (in press)
5. Blackburn TH, Hall POJ, Hulth S et al (1996) Organic-N loss by efflux and burial associated with a low efflux of inorganic N and with nitrate assimilation in Arctic sediments, Svalbard (Norway). Mar Ecol Prog Ser 141:283–293
6. Booth C (1978) In: Booth C (ed) Introduction to general methods. Methods in Microbiology. Academic Press, New York, pp 57–91
7. Chintalapati S, Kiran MD, Shivaji S (2004) Role of membrane lipid fatty acids in cold adaptation. Cell Mol Biol (Noisy-le-grand) 50:631–642
8. Elverhøi A, Liestøl O, Nagy J (1980) Glacial erosion, sedimentation and microfauna in the inner part of Kongsfjorden, Spitsbergen. Norsk Polarinstitutt Skrifter 172:33–60
9. Feller G, Gerday C (2003) Psychrophilic enzymes: hot topics in cold adaptation. Nat Rev Microbiol 1:200–208
10. Fong NJ, Burgess ML, Barrow KD et al (2001) Carotenoid accumulation in the psychrotrophic bacterium *Arthrobacter agilis* in response to thermal and salt stress. Appl Microbiol Biotechnol 56:750–756
11. Good IJ (1954) The population frequencies of species and the estimation of population parameters. Biometrica 40:237–264
12. Hao MV, Komagata K (1985) A new species of *Planococcus*, *P.* kocurii isolated from fish, frozen foods, and fish curing brine. J Gen Appl Microbiol 31:441–455
13. Hop H, Borgá K, Gabrielsen GW et al (2002) The marine ecosystem of Kongsfjorden, Svalbard. Polar Research 21:167–208

14. Jagannadham MV, Rao VJ, Shivaji S (1991) The major carotenoid pigment of a psychrotrophic *Micrococcus roseus* strain: purification, structure, and interaction with synthetic membranes. J Bacteriol 173:7911–7917
15. Jagannadham MV, Chattopadhyay MK, Shivaji S (1996) The major carotenoid pigment of a psychrotrophic Micrococcus roseus strain: fluorescence properties of the pigment and its binding to membranes. Biochem Biophys Res Commun 1996(220):724–728
16. Jankowska K, Wlodarska-Kowalczuk M, Wieczorek P (2005) Abundance and biomass of bacteria in two Arctic glacial fjords. Pol Polar Res/Pol Badania Polarne 26:77–84
17. Jorgensen LL, Gulliksen B (2001) Rocky bottom fauna in arctic Kongsfjord (Svalbard) studied by means of suction sampling and photography. Polar Biol 24:113–121
18. Knoblauch C, Jørgensen BB, Harder J (1999) Community size and metabolic rates of psychrophilic sulfate-reducing bacteria in Arctic marine sediments. Appl Environ Microbiol 65:4230–4233
19. Kostka JE, Thamdrup B, Glud RN, Canfield DE (1999) Rates and pathways of carbon oxidation in permanently cold Arctic sediments. Mar Ecol Prog Ser 180:7–21
20. Lane DJ (1991) 16S/23S rRNA sequencing. In: Stackebrandt E, Good Fellow M (eds) Nucleic acid techniques in bacterial systematics. Wiley, Chichester, pp 115–175
21. Lefauconnier B, Hagen JO, Rudant JP (1994) Flow speed and calving rate of Kongsbreen glacier, Svalbard, using SPOT images. Pol Res 10:56–65
22. Magurran AE (1996) Ecological diversity and its measurement. Chapman and Hall, London, pp 274–286
23. Mueller DR, Vincent WF, Bonilla S et al (2005) Extremotrophs, extremophiles and broadband pigmentation strategies in a high arctic ice shelf ecosystem. FEMS Microbiol Ecol 53:73–87
24. Nishida I, Murata N (1996) Chilling sensitivity in plants and cyanobacteria: the crucial contribution of membrane lipids. Annu Rev Plant Physiol Plant Mol Biol 47:541–568
25. Priest FG (1977) Extracellular enzyme synthesis in the genus *Bacillus*. Bacteriol Rev 41:711–753
26. Sahm K, Berninger U (1998) Abundance, vertical distribution, and community structure of benthic prokaryotes from permanently cold marine sediments (Svalbard, Arctic Ocean). Mar Ecol Progr Ser 165:71–80
27. Sahm K, Knoblauch C, Amann R (1999) Phylogenetic affiliation and quantification of psychrophilic sulfate-reducing isolates in marine Arctic sediments. Appl Environ Microbiol 65:3976–3981
28. Shivaji S, Gupta P, Chaturvedi P, Suresh K, Delille D (2005) *Marinobacter maritimus* sp. nov., a psychrotolerant strain isolated from sea water off the subantarctic Kerguelen islands. Int J Syst Evol Microbiol 55:1453–1456
29. Shivaji S, Chaturvedi P, Suresh K et al (2006) *Bacillus aeriu*s sp. nov., *Bacillus aerophilus* sp. nov., *Bacillus stratosphericus* sp. nov. and *Bacillus altitudinis* sp. nov., isolated from cryogenic tubes used for collecting air samples from high altitudes. Int J Syst Evol Microbiol 56:1465–1473
30. Shivaji S, Bhadra B, Rao RS et al (2007) *Microbacterium indicum* sp. nov., isolated from a deep-sea sediment sample from the Chagos Trench, Indian Ocean. Int J Syst Evol Microbiol 57:1819–1822
31. Shivaji S, Kiran MD, Chintalapati S (2007) Perception and transduction of low temperature in bacteria. In: Gerday C, Glansdorff N (eds) Physiology and biochemistry of extremophiles. Washington, DC, ASM Press, pp 194–207
32. Svendsen H, Beszczuńska-Moller A, Hagen JO et al (2002) Physical environment of Kongsfjorden–Krossfjorden, an Arctic fjord system in Svalbard. Polar Res 21:133–166
33. Thompson JD, Higgins DG, Gibson TJ et al (1997) The CLUSTAL_X windows interface: flexible strategies for multiple sequence alignment aided by quality analysis tools. Nucleic Acids Res 25:4876–4882
34. Väätänen P (1977) Effects of composition of substrate and inoculation technique on plate counts of bacteria in the Northern Baltic Sea. J Appl Bacteriol 42:431–443
35. Van Trappen S, Vandecandelaere I, Mergaert J, Swings J (2004) *Flavobacterium degerlachei* sp. nov., *Flavobacterium frigoris* sp. nov. and *Flavobacterium micromati* sp. nov., novel psychrophilic bacteria isolated from microbial mats in Antarctic lakes. Int J Syst Evol Microbiol 54:85–92
36. Weiner RM, Segall AM, Colwell RR (1985) Characterization of marine bacteria associated with *Crassostrea virginica* (Eastern oyster). Appl Environ Microbiol 49:83–90
37. Wessels H, Hagen W, Molis M, Wiencke C, Karsten U (2006) Intra- and interspecific differences in palatability of Arctic macroalgae from Kongsfjorden (Spitsbergen) for two benthic sympatric invertebrates. J Exp Mar Biol Ecol 329:20–33
38. Wlodarska-Kowalczuk M, Weslawski JM (2001) Impact of climate warming on Arctic benthic biodiversity: a case study of two Arctic glacial bays. Climate Res 18:127–132
39. Zobell CE (1946) Marine microbiology. Chronica Botanica Co., Waltham, MA